

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q22-2056

Blythe 100 kV
March 02, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2056. The Interconnection Customer proposes a 125.2 MW solar photovoltaic facility interconnecting at the Blythe 100 kV substation in the Illinois area. The request was received on April 06, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 497 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$16,566,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Matteson / Hoosier Fairview 100 kV Circuit 1	Matteson / Hoosier Fairview 100 kV #2	122.2	140	6.4
Matteson / Hoosier Fairview 100 kV Cir. 2	Matteson / Hoosier Fairview 100 kV Line 1	122.2	140	6.4
Spearfish Hydro / Hoosier Fairview 100 kV Line 1	Spearfish Hydro / Hoosier Fairview 100 kV #2	105.2	213	11.6
Hoosier Fairview / Spearfish Hydro 100 kV #2	Hoosier Fairview / Spearfish Hydro Substation 100 kV Line 1	105.2	213	11.6
Lincoln / Blythe 230 kV	S Boston / Blythe 230 kV Line	105.1	617	53.8
Sweatt / Arch Smith 100 kV Line 1	Arch Smith / Swatt 100 kV Line 2	104.4	451	20.2
Sweatt / Arch Smith 100 kV Line 2	Sweatt / Arch Smih 100 kV Cir. 1	104.4	451	20.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Matteson / Hoosier Fairview 100 kV Circuit 1	Rebuild line with higher-capacity conductor	\$8,862,000	100% (\$8,862,000)
Matteson / Hoosier Fairview 100 kV Cir. 2	Rebuild line with higher-capacity conductor	\$6,970,000	100% (\$6,970,000)
Spearfish Hydro / Hoosier Fairview 100 kV Line 1	Reconductor line	\$442,000	8% (\$34,000)
Hoosier Fairview / Spearfish Hydro 100 kV #2	Reconductor line	\$766,000	8% (\$58,000)
Lincoln / Blythe 230 kV	Reconductor line	\$6,731,000	8% (\$550,000)
Sweatt Substation / Arch Smith 100 kV Ckt 1	Reconductor line	\$494,000	8% (\$42,000)
Sweatt / Arch Smith 100 kV Line 2	Reconductor line	\$585,000	8% (\$50,000)
	Total allocated to project		\$16,566,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.